

PEOPLE COUNTER

Using An IR Sensor

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A people counter enables real-time occupancy monitoring, making it ideal to check occupancy in offices, factories, libraries, meeting halls, and other public spaces. The system presented here is a simple, reliable, and cost-effective people-counting solution based on an IR sensor and CD4026 decade counter ICs. Unlike microcontroller-based alternatives, this design is easy to construct, troubleshoot, and maintain, eliminating the need for programming. It is therefore well suited for beginners, educational demonstrations, and practical installations requiring a low-cost, dependable people counter. The author's prototype on a breadboard is shown in Fig. 1.

Circuit and working

Fig. 2 shows the circuit diagram of the people counter using an IR sensor module. It is built around two CD4026 decade counter/7-segment driver ICs (IC1 and IC2), two common-cathode 7-segment displays (DIS1 and DIS2, such as LTS6980HR or LTS543), and a few other components.

The two CD4026 ICs are cascaded to count and display numbers from 00 to 99. An IR sensor module detects the movement of a person passing through the entrance, and the count is displayed on the two 7-segment displays. Other components include a 2.2k Ω resistor (R1) for stable operation, 330 Ω resistors (R2 and R3) for display current limiting, and a push-to-on switch (S1) for resetting the counter to zero.

IC1 drives the units digit display (DIS1), while its carry-out output

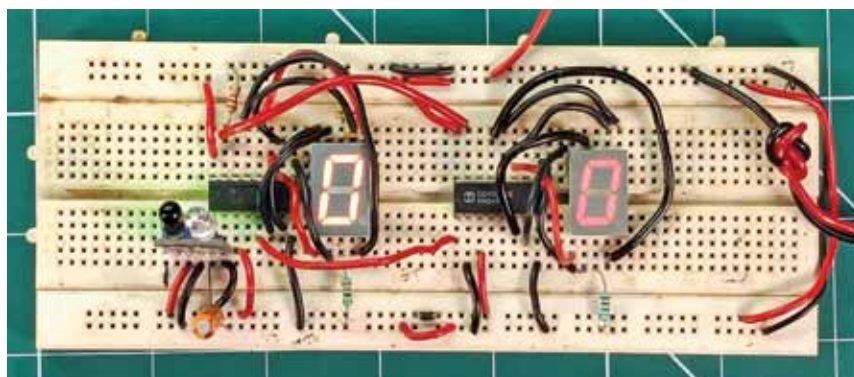


Fig. 1: Author's prototype

is connected to the clock input of the second IC (IC2), which drives the tens digit display (DIS2). This cascading arrangement enables the circuit to count from 00 to 99. A reset switch (S1) simultaneously clears both counters and resets the display to zero.

Proper grounding and the use of current-limiting resistors ensure reliable operation. The sensitivity of the IR sensor can be adjusted using its onboard preset to achieve accurate detection and minimise false triggering. The entire circuit operates off a regulated 5V DC power supply or adaptor. The output of the 5V DC adaptor is connected to CON1 to power the circuit.

The people counter operates by detecting and counting digital pulses generated by the IR sensor. The IR sensor continuously emits infrared radiation and monitors its reflection. When a person passes in front of the sensor, the IR beam is interrupted, generating a digital pulse at its output. This pulse is applied to the clock input of the first CD4026 (IC1), causing the count to increment by one. Since the CD4026 incorporates

a 7-segment decoder/driver, it directly drives the displays (DIS1 and DIS2). The maximum count displayed by DIS1 and DIS2 is 99. To extend the count to three or four digits, you may cascade three or four CD4026 ICs, respectively.

Construction and testing

An actual-size, single-sided PCB layout for the people counter is shown in Fig. 3, and its component layout is shown in Fig. 4. After assembling the circuit on the designed PCB, mount it inside a suitable enclosure with proper openings for the displays, the reset switch, the connectors, and the IR sensor. Ensure that all power supply pins are connected correctly to +5V and ground. Alternatively, the circuit may be assembled on a general-purpose PCB if the designed PCB is unavailable.

Mount the displays DIS1 and DIS2, along with the reset switch (S1), on the front panel of the enclosure. Position the IR sensor module at a suitable height and angle so that it reliably detects a person crossing its path. The sensor output is connected to the clock